

Experimental Demonstration of Resonant Adiabatic Writing and Computing in Ferroelectric Capacitive Memory Array for Energy-Efficient Edge AI

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Abstract—In this work, a novel resonant adiabatic (RA) capacitive computing-in-memory (C-CIM) with non-volatility is proposed and experimentally demonstrated based on 3D stackable ferroelectric (FE) capacitor (FeCap) array, enabling both writing and multiply-accumulate (MAC) computing with record-high energy efficiency. For the first time, by leveraging the capacitive nature of hafnia-based FeCap, both capacitance state writing and non-destructive readout for computing are implemented with incorporated resonant energy recovery and adiabatic charging in FeCap based C-CIM. The experimentally measured power consumption indicates that the proposed ferroelectric RA C-CIM can significantly reduce dynamic power loss inherent in conventional pulse-based C-CIM, along with suppressing resistive heating dissipation. Further co-designing FeCap-based RA C-CIM with layer normalization algorithm, high-accuracy pattern recognition is demonstrated with $35\times/8.95\times$ energy efficiency improvement compared with conventional resistive and capacitive CIM respectively, providing a promising solution for energy-efficient edge AI.

I. INTRODUCTION

Computing-in-memory (CIM) architectures based on 3D stacking memory technology have been widely studied for deep neural network (DNN) accelerator, due to significantly enhanced compute density and reduced data transfer, while suffering from potential heat dissipation problems [1] (Fig. 1a). For hardware implementation, capacitive memory devices [2-9] can eliminate the irreversible Joule heat dissipation during writing and computing in resistive memory (R-CIM) (Fig. 1b). Moreover, benefiting from its inherent insulation nature, adiabatic charge recycling technique with reversible energy recovery has great potential to be exploited in capacitive memory based CIM (C-CIM) (Fig. 1c) for further minimizing dynamic power consumption of charging and discharging, which fundamentally limits the energy efficiency boost of conventional clock pulse-based C-CIM array [3-4] (Fig. 1d). However, there has been few demonstrations of adiabatic techniques on C-CIM based on programmable capacitive memory devices. Recently, an adiabatic C-CIM with charge-trapping-based programable capacitor has been conceptually reported with simulation [4], though it suffers from high write power consumption due to charge injection. In recent years, Hafnia-based ferroelectric (FE) devices have garnered significant attention for CIM applications [5-8] due to their high CMOS compatibility, good scalability, and low-energy field-driven write capabilities.

In this work, a novel C-CIM with both resonant adiabatic writing and computing is proposed and experimentally demonstrated based on 3D stackable FE capacitor (FeCap) array for the first time. By leveraging the capacitive write/read

characteristics of hafnia-based FeCap, resonant energy recovery and adiabatic charging are incorporated, expanding the potential for extreme reduction of power loss of C-CIM. Co-optimizing the design and normalization algorithm, high-accuracy pattern recognition is demonstrated with a record-high energy efficiency for weight update and computing.

II. DESIGN AND EXPERIMENTS OF RESONANT ADIABATIC C-CIM BASED ON FE CAP MEMORY ARRAY

A. Ferroelectric resonant adiabatic (RA) C-CIM architecture

The proposed resonant adiabatic (RA) C-CIM is shown in Fig. 2. The 3D stackable FeCap enables capacitive weight (w) storage and programing without static DC current [6-8]. Resonant adiabatic drivers (Fig. 2c) generate write voltage (V_W) and non-destructive read voltage (V_R), incorporating an inductor for energy recovery and two anti-phase switches for resonance connection (S_R) and dissipated energy compensation (S_C). The decoders of word-line (WL) and bit-line (BL) route V_W and V_R . During writing or computing stage, periodic V_W (or V_R) with harmonic form is generated through the intrinsic resonance between an inductor and FeCaps, in contrast to clock-controlled pulse supplies in conventional C-CIM [7]. This design synergistically leverages resonant energy recovery and adiabatic charging (Fig. 2a). **Resonant energy-recovery:** The energy used to charge FeCaps can be transferred to inductor after one write (/compute) operation, subsequently recycled for the next cycle. Compared with conventional pulse-based C-CIM, this approach allows to recapture the discharging energy losses when switching decoders between consecutive cycles. **Adiabatic charging:** the FeCaps are charged through a ramping harmonic voltage, rather than the pulse-based hard charging in conventional C-CIM. This approach reduces Joule heating dissipation in the resistive components due to lower voltage in the resistive path during charging. Therefore, the resonant adiabatic FeCap C-CIM optimizes dynamic power and resistive dissipation during both writing and multiply-accumulate (MAC) computing stages, and is expected to reduce energy consumption to ultralow level.

B. FeCap based resonant adiabatic (RA) C-CIM system

Based on the above design, a resonant adiabatic C-CIM system (Fig. 3) is constructed. The circuit modules, including 4-channel RA drivers, decoders, integrators, are developed on a PCB (Fig. 3b). The capacitive weight matrix (W) is realized by using our previously fabricated 3D-stackable cross-point FeCap array, which features high disturb immunity optimized by electrodes and interfacial engineering [10] (Fig. 3c-f). The FeCap, composed of a TiN/HZO/W/Ti/Pt stack, exhibits typical FE P-V and small-signal C-V characteristics under relatively large sweep voltage range (Fig. 3g). The asymmetric electrode design further optimizes the capacitance memory

window (C_{MW}) around 0V readout bias. By applying V_W to set the FE polarization direction to up/down state, the FeCap can be adjusted to high/low capacitance state (HCS/LCS) (Fig. 3h), which exhibits non-destructive C-V readout under relatively small V_R range from -0.1V to 0.1V. It shows good retention and writing endurance (Fig. 3ij).

III. RESONANT ADIABATIC WEIGHT WRITING FOR FECAP-BASED C-CIM

A. Resonant adiabatic writing operation for FeCap array

The $V_W/3$ write scheme is used for 1k-bit FeCap cross-point array without cell-wise selectors. This scheme operates the selected cell with V_W amplitude and brings a disturb voltage of $\pm V_W/3$ to unselected cell, resulting in FE polarization back-switching and C_{MW} decrease (Fig. 4ab). Due to the electric field and time dependent multi-domain FE switching characteristic of HZO layer, the C_{MW} before disturb generally increases with increased V_W and write period (t_W), which however, also causes more disturbance (Fig. 4cd). Consequently, C_{MW} after disturbance is not completely monotonic with V_W and t_W , showing multiple local optimization conditions (Fig. 4e). Further considering the optimization of write power, $V_W=2.1V$ and $t_W=1\mu s$ are used for resonant adiabatic writing by selecting an appropriate inductor to achieve the desired resonance frequency.

The measured timing diagram of resonant adiabatic writing for FeCap array is shown in Fig. 5ab. Three voltage harmonics (V_W , $V_W/3$, $2V_W/3$) are generated through resonance between the inductor and on-chip FeCaps along with a bypass capacitor, and applied to WLs and BLs according to the decoders. The capacitance and FE-polarization state after writing operation are shown in Fig. 5c-e, indicating that the capacitive weights with complementary form are successfully written to the cross-point FeCap array even with disturbed harmonic cycles.

B. Write power of resonant adiabatic FeCap-based C-CIM

The power consumption of above writing process for FeCap array is further experimentally measured using the proposed resonant adiabatic schemes and conventional pulse-based scheme, respectively (Fig. 6). Consistent with the analysis of optimized write conditions, relatively low V_W of 2.1V can save lots of write power consumption while keep functional C_{MW} (Fig. 6a). Compared with power of pulse-based write, the total power of RA writing for FeCap array is improved by $3.24\times$ under continuously switching the decoders, even with the presence of PCB parasitic in RA drivers (Fig. 6b). Moreover, the dynamic power for writing FeCap array obtained by comparing power consumption under decoder switching and non-switching, showing significant advantages ($83.6\times$) with the proposed RA writing scheme. It is due to that the charging energy being temporarily stored in the FeCap during the writing stage, is subsequently transferred to inductor through resonance. This energy can be used for the next write cycle, resulting in improved energy efficiency for weight update in FeCap-C-CIM.

IV. FECAP-BASED RESONANT ADIABATIC MAC OF C-CIM

A. FeCap-based resonant adiabatic (RA) MAC computing

After RA writing operation, the capacitive weight matrix is

stored in FeCap array as HCS/LCS (Fig. 7a). For MAC of C-CIM, the bit-wise input x vector is implemented with row-wise switches, which controls WL to connect to V_R (if $x=1$) or GND (if $x=0$). The weighted induced charge is summed through the column-wise integrators, where V_{out} is proportional to MAC results. Instead of using DC V_R supply in the conventional C-CIM, RA driver is used for V_R generation through resonance between inductor and weighted FeCap array. Moreover, complementary FeCaps are employed for differential weight storage instead of a single FeCap. This ensures the total load capacitance and resonant period remain independent with weight data ("1"/"-1"), stabilizing V_R and operating frequency (Fig. 7bc). For FeCap in HCS/LCS, the input V_R causes V_{out} with high/low amplitude (Fig. 7d), and the differential voltage represents $x_i \cdot w_{ij}$. The timing diagram of FeCap-based RA C-CIM is experimentally captured (Fig. 7e). The rhythm of MAC is controlled by S_R and S_C , and the C_{MW} of differential w is stable for cycling readout (Fig. 8).

B. Power of FeCap-based resonant adiabatic MAC computing

Furthermore, the power consumption (P_{RA}) of supply for above RA MAC process in FeCap array is measured. It demonstrates advantage compared with conventional pulse-based scheme (P_{con}) (Fig. 9a), similar to RA writing process due to capacitive nature of FeCap. Moreover, as the toggle rate of input x increases, P_{con} rise rapidly due to that more charge is shunted to GND, resulting in serious dynamic power dissipation (Fig. 9b). In contrary, benefitting from charge recycling, P_{RA} is stable, and the integrated charge shows linearity with x -dependent MAC results (Fig. 10).

V. FECAP-BASED RESONANT ADIABATIC C-CIM MACRO

Based on the proposed resonant adiabatic FeCap C-CIM with energy recovery and heating suppression during both writing and computing, the neural network for pattern recognition is demonstrated. The parasitic in PCB are excluded through circuit simulation with extracted FeCap characteristics. The layer normalization algorithm (Fig. 11), which makes the distribution of x symmetrical, is applied to the proposed RA C-CIM for stabilized loaded capacitance and resonance period with various input patterns, while maintaining high accuracy. In addition, the energy efficiency of FeCap-based RA C-CIM is evaluated and compared, showing reduced both dynamic and resistive power dissipation due to charge recycling and adiabatic charging (Fig. 12). The subarray MAC energy can be reduced by $107\times$ (Fig. 13), and the energy efficiency is evaluated with consideration of the ADC [13], showing significantly improvement compared with conventional R-CIM ($35\times$) and C-CIM ($8.95\times$).

VI. SUMMARY

This work reports the first experimental demonstration of resonant adiabatic writing and computing for C-CIM based on 3D-stackable FeCap array, enabling significantly reduced dynamic power consumption and resistive dissipation for weight update and MAC operation (Table 1). Further co-optimizing the proposed design with layer-wise normalization algorithm, high-accuracy pattern recognition is demonstrated with record-high energy efficiency, highlighting its great potential for edge AI application.

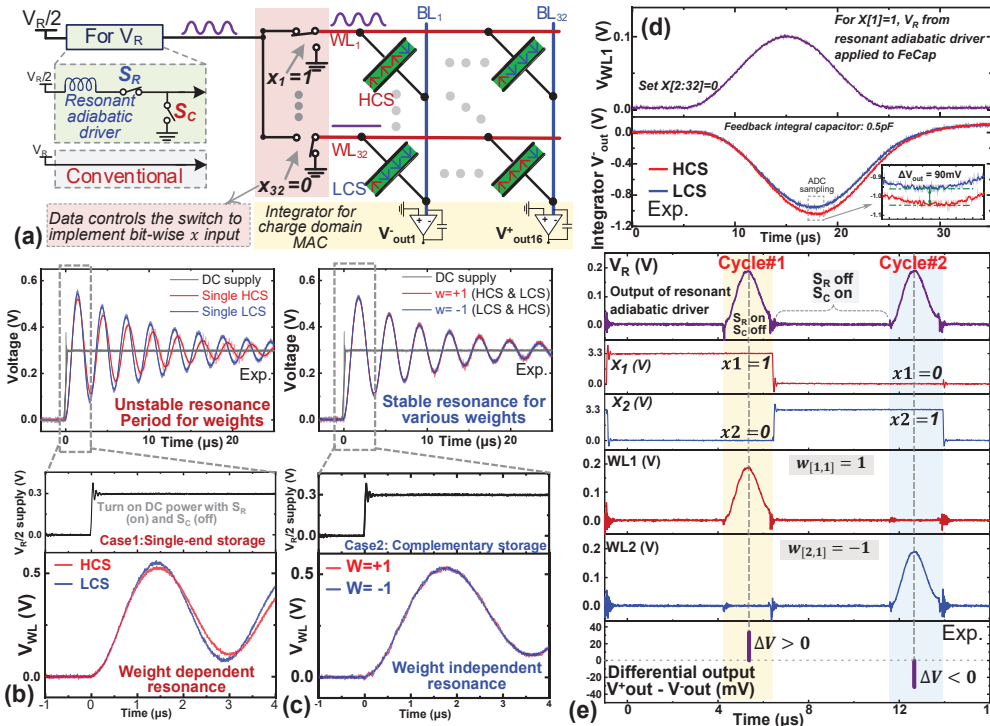


Fig. 7 Measured resonant adiabatic MAC computing process of FeCap array-based C-CIM. (a) The circuit schematic for RA MAC operation. The capacitive weight matrix is written into FeCap array via RA writing. The resonance process (S_R -on and S_C -off) between weighted FeCap and off-chip inductor in RA driver based on (b) single-end and (c) complementary encoding for weight, indicating that complementary weight can stabilize loading capacitance and resonance period. (d) The output voltage (V_{out}) of integrator for multiplication in each FeCap. (e) The timing diagram when MAC computing, where $X[3:32]=0$. The differential voltage of complementary BLs indicating the correct MAC results.

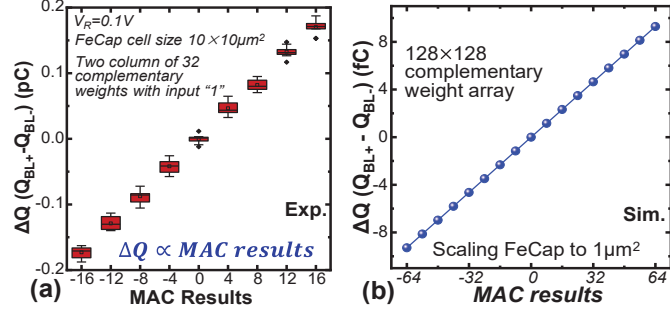


Fig. 10 (a) Measured MAC results of a column of FeCap weights by extracting the summed charge through integration, indicating the linear computing. (b) Simulated FeCap-based resonant adiabatic MAC in HSPICE.

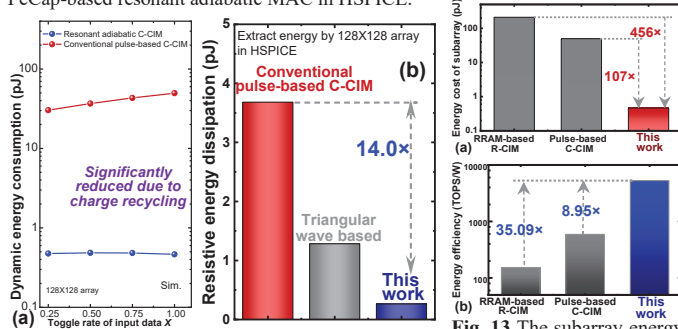


Fig. 12 (a) Energy consumption of dynamic charging for MAC computing. (b) Resistive energy consumption extracted from the parasitic wire resistance is reduced due to ramp adiabatic charging.

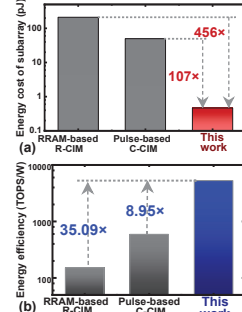


Fig. 13 The subarray energy cost and (b) the energy efficiency of conventional R-CIM [7,11], C-CIM [8], and the proposed FE RA C-CIM.

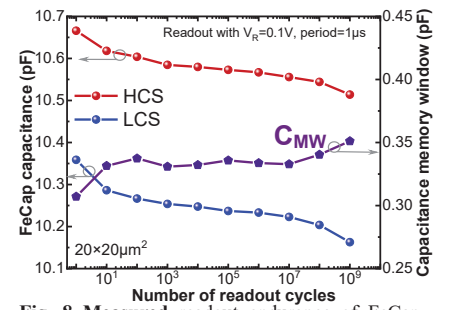


Fig. 8 Measured readout endurance of FeCap-based capacitive weight for continuous MAC computing. The functional CMW is maintained even with 10^9 RA computing cycle.

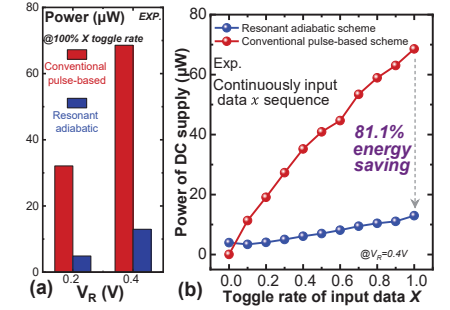


Fig. 9 Measured power consumption of MAC process in FeCap-based RA C-CIM. (a) The comparison with conventional pulse-based scheme. (b) The power consumption when x vector changes during continuous computing cycles, where the frequency of charging is proportional to toggle rate.

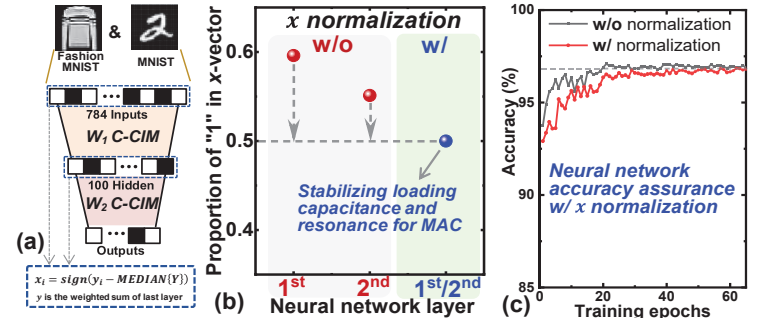


Fig. 11 Co-design of layer-wise activation normalization algorithm and the proposed FeCap-based resonant adiabatic C-CIM. (a) The schematic diagram of x normalization and the artificial neural network structure. (b) Half of x are activated for stabilized load capacitance and resonance period. (c) Learning process of FeCap-based RA C-CIM with normalization.

Table 1 Benchmark of proposed FeCap-based resonant adiabatic writing and computing enhanced C-CIM in this work with other reported solutions.

	[11]	[13]	[4]	[7-9]	This work
Memory type	RRAM	FeRAM (PZT)	Charge-trapping MOS Cap	FeCap	FeCap (HZO)
Storage & Readout principle	Resistance	FE polarization switching	Capacitance memory window (C_{MW})		C_{MW}
CIM type	R-CIM	/	C-CIM	C-CIM	C-CIM
Weight write scheme	Current	Adiabatic	Pulse-based	Conventional pulse-based	Resonant adiabatic
MAC scheme	Current	/	Adiabatic (Sim)		
Writing energy (pJ/bit)	4.5	0.99	high (>5V, >1ms)	0.82 ^(a)	0.25pJ/bit ^(c)
Subarray energy	210.7pJ ^(b)	/	95.6pJ	49.7pJ ^(b)	0.46pJ
Energy efficiency (TOPS/W)	151.4 ^(bc)	/	3452.6	591.6 ^(c)	5298 TOPS/W

(a) Extracted from experimental measurement. (b) Calculated in HSPICE based on same array size with respectively proposed weight cell characteristics. (c) The energy of ADC is extracted from [12].

Acknowledgements: This work was supported by NSFC (61927901, 62374009), 111 Project (B18001) and China Postdoctoral Science Foundation (2024M750098, GZC20240026). **Reference:** [1] A. Kaul, *et al.*, *IEDM*, p. 259-262, 2020. [2] Z. Zhou, *et al.*, *VLSI*, p. 357-358, 2022. [3] R. Karakiewicz, *et al.*, *JSSC*, p. 2573-2584, 2007. [4] K. U. Demasius, *et al.*, *Nat. Electron.*, p. 748-756, 2021. [5] M. S. Song, *et al.*, *IEDM*, p. 12.2.1-12.2.4, 2023. [6] Q. Zheng, *et al.*, *EDL*, p. 1309-1312, 2019. [7] Y. C. Luo, *et al.*, *TCAS-II*, p. 784-788, 2021. [8] S. Mukherjee, *et al.*, *IEDM*, p. 11.4.1-11.4.4, 2023. [9] W. Xu, *et al.*, *EDL*, p. 24-27, 2024. [10] Z. Fu, *et al.*, *IEDM*, p. 11.3.1-11.3.4, 2023. [11] P. Yao, *et al.*, *Nature*, p. 641-646, 2020. [12] L. Xu, *et al.*, *DAC*, p. 115-120, 2022. [13] S. Jeloka, *et al.*, *VLSI*, p. 85-86, 2018.